

Fig. 3: An example of good routing (left) and bad routing (right)



Fig. 6: Adding chamfer to 'T' junctions eliminates 90° angles

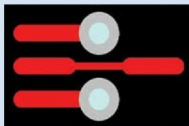


Fig. 4: Necking down

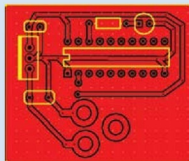


Fig. 5: PCB layout looks like this if you leave rest of the space covered by copper, leaving clearance beside each line



Fig. 7: Screenshot of the window for printing the PCB layout

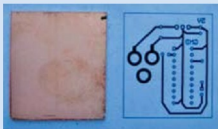


Fig. 8: Printout of the PCB layout

2. Rotate only the component by 90°
3. It is the component and symbol manager. All the components can be found here
4. Options. You can change the colour and dimensions of the screen as well as the grid spacing and other options
5. Previous zoom. It resumes the previous screen shape
6. Zoom to fit
7. Zoom out
8. Zoom in
9. Undo
10. Save
11. Open
12. General-purpose pointer, select and moving tool
13. Zoom into selected area
14. Place a component. It will not open the library window
15. Place a signal or symbol level,

like the symbols of power supply and ground

16. Place a wire. All the connections are made using this icon

17. Place a corner in the wire, i.e., break a continuous line into two

18. Disconnect a wire from a component and connect it to the other

19. Make a rectangle for a new customised component. You can make any component which is not present in the library

20. Add a line into a customised component

21. Add a circle or arc in a new component

22. Place a new pin in a component

23. Place text

24. Snap to grid. By keeping this option 'on,' the wires will move and



Fig. 9: A full-size copper-clad board



Fig. 10: Paper put on the board with the printed side facing the copper side (left) and affixed using cello tape (right)



Fig. 11: Set the temperature of your electric iron to maximum

make angles at specific gap. This makes it easy to draw straight lines and designs. By keeping this option 'off,' you will be able to make high-precision angles in the circuit

## Designing the PCB

While designing a PCB, try to make it as compact as possible.

Fig. 2 shows the PCB of the circuit (shown in Fig. 1) designed in ExpressPCB. Yellow lines show the silkscreen layer (component layout), red lines show the top copper layer and green portion is the bottom copper layer.

**Screenshot description.** Icons 1 to 31 in Fig. 2 allow you to perform following tasks:

1. Rotate the component as well as the text by 90°